

Review 2.1

B. Tech. Project (19ECE495)

Batch ID: **070**

Mutual coupling reduction in MIMO Patch Antenna Array for Wi-Fi and Bluetooth applications

CB.EN.U4ECE22225 - J Rohith

CB.EN.U4ECE22262 - Peddakudala Yashwanth

CB.EN.U4ECE22213 - Chundupalle Tej Siddanth

MENTOR: Ms. V. MEKALADEVI

Dept. of Elec. and Commn. Engg.
Amrita School of Engineering, Coimbatore

OBJECTIVE:

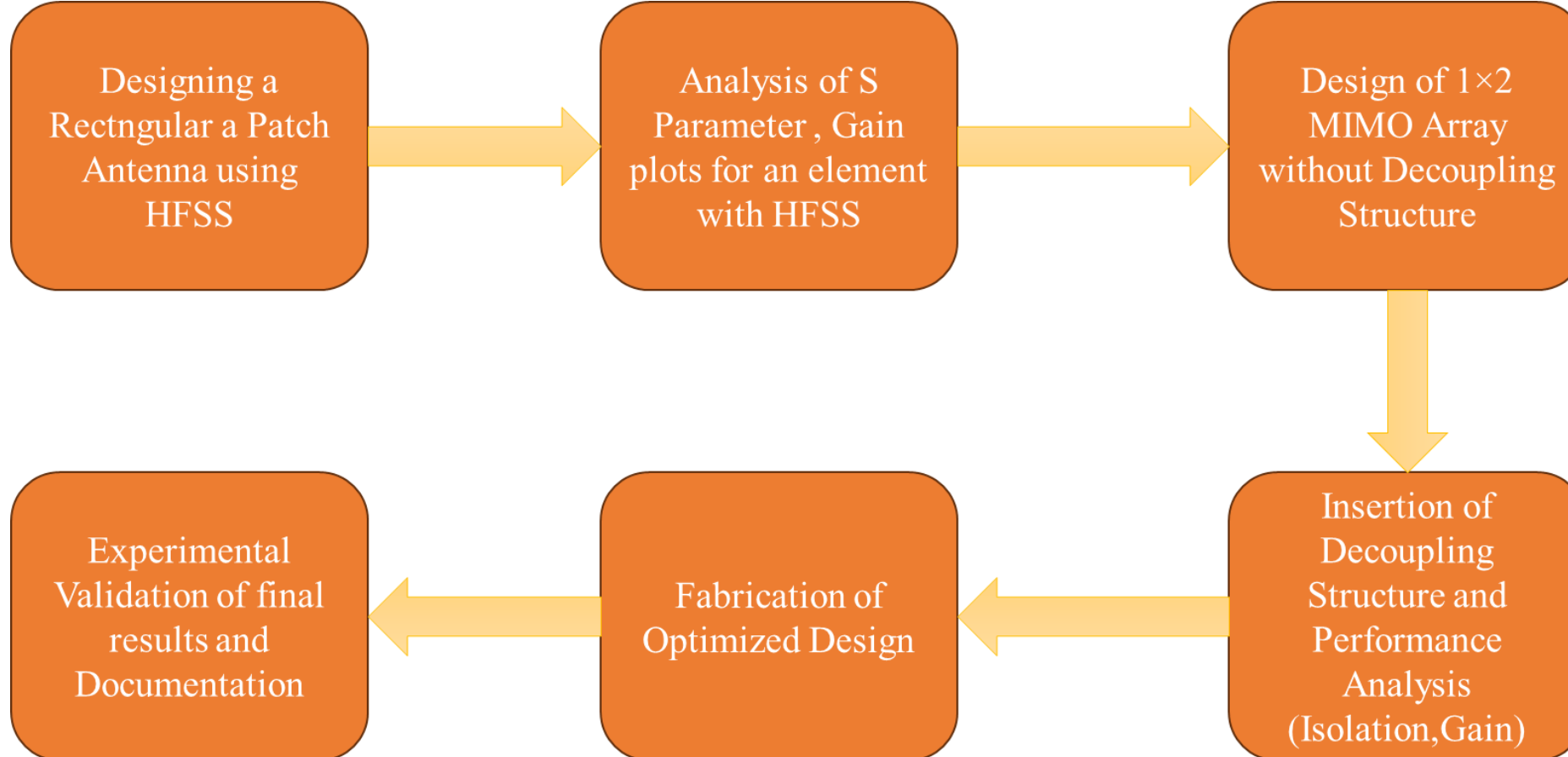
- Design and analysis of base antenna at 2.4 GHz.
- Design and analysis of Patch antenna array at 2.4 GHz.
- Design and analysis of Patch antenna array with decoupling structures.
- Prototype and Measurement.

Performance Targets:

- Isolation:** > 20 dB between antenna elements
- Element Spacing:** $< \lambda/2$ for compact design
- Operating Frequency:** 2.4 GHz (Wi-Fi and Bluetooth bands)

Review	Month	Milestone
1.1	June-July 2025	Literature Survey of Decoupling Structures.
1.2	August – September 2025	Design and Analysis of Base Antenna at 2.4 GHZ.
1.3	October-November 2025	Design and Analysis of Patch Antenna Array at 2.4 GHZ.
2.1	December 2025-January 2026	Design and Analysis of Patch Antenna Array with De-Coupling Structure at 2.4 GHZ.
2.2	February-March 2026	Prototype and Measurement

METHODOLOGY:



METHODOLOGY:

□ Patch Antenna Dimension Calculation:

- As the frequency of the application is 2.4 GHz, considering Rogers RT duroid 5880 ($\epsilon_r = 2.2$) is a suitable substrate, the basic patch antenna dimensions are determined through mathematical calculations.
- Designed for 2.4 GHz rectangular patch on Rogers RT duroid 5880 ($\epsilon_r = 2.2$) with $h = 1.6$ mm.

Patch Width(W):

$$W = \frac{c}{2f_0} \sqrt{\frac{2}{\epsilon_r + 1}} \quad \dots(1)$$

$$W = \frac{3 \times 10^8}{2 \cdot 2.4 \times 10^9} \sqrt{\frac{2}{3.2}} \Rightarrow W = 49.38 \text{ mm}$$

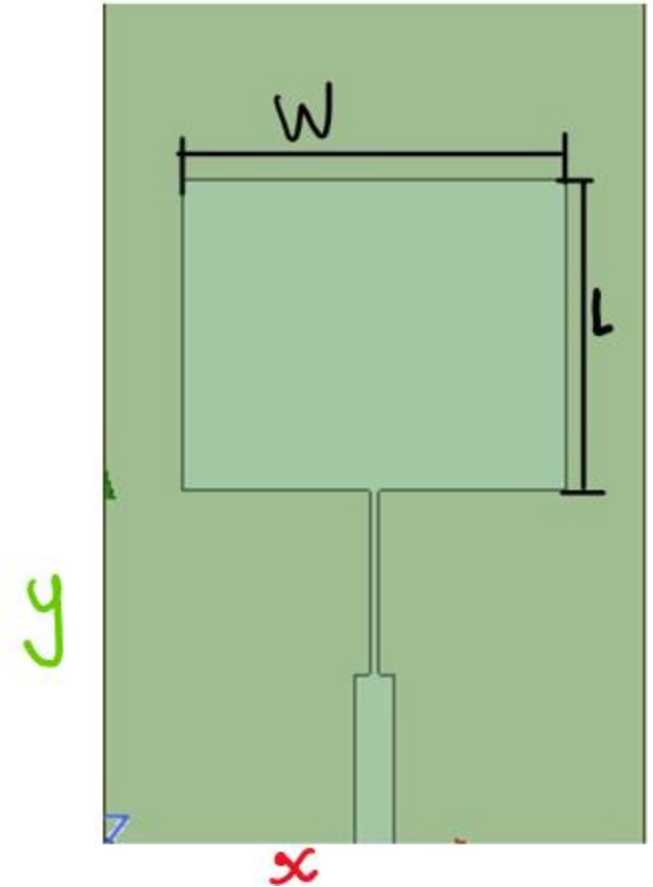


Figure 1. Simulated model of a unit element of an Array

□ Patch Antenna Dimension Calculation:

Effective Permittivity ϵ_{eff}

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{12h}{W} \right)^{-1/2} \quad \dots(2)$$

$$\epsilon_{\text{eff}} = \frac{2.2 + 1}{2} + \frac{1.2}{2} \cdot \frac{1}{\sqrt{1 + \frac{12(1.6)}{38}}} \Rightarrow \epsilon_{\text{eff}} \approx 2.03$$

Fringing extension ΔL

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{\text{eff}} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{\text{eff}} - 0.258) \left(\frac{W}{h} + 0.8 \right)} \quad \dots(3)$$

$$\Delta L = 0.412(1.6) \cdot \frac{(2.03 + 0.3)(23.75 + 0.264)}{(2.03 - 0.258)(23.75 + 0.8)} \Rightarrow \Delta L \approx 0.74 \text{mm}$$

□ Patch Antenna Dimension Calculation:

Effective length(L_{eff}):

$$L_{\text{eff}} = \frac{c}{2f_0\sqrt{\epsilon_{\text{eff}}}} \quad \dots(4)$$

$$L_{\text{eff}} = \frac{3 \times 10^8}{2 \times 2.4 \times 10^9 \sqrt{2.03}} \Rightarrow L_{\text{eff}} = 30.2\text{mm}$$

Physical length of patch:

$$L = L_{\text{eff}} - 2\Delta L \quad \dots(5)$$

$$L = 30.2 - 1.48 = 28.7\text{mm}$$

Results:

- Analysis of the S-parameter plot for an element with HFSS.

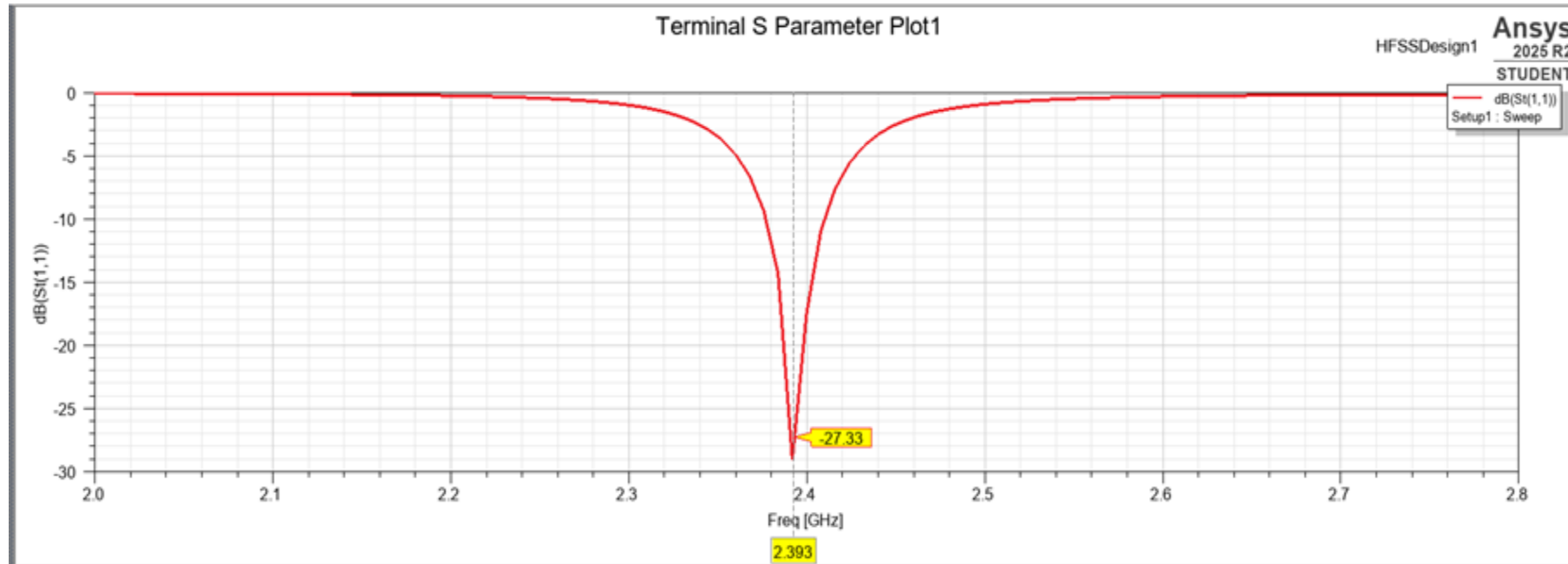


Figure 2. Simulated S-parameter of a unit element

METHODOLOGY:

Design of 1×2 Patch antenna Array without Decoupling Structure:

- The spacing between the centres of the two patches is chosen to be more than $0.5 \lambda_0$ at 2.4 GHz.
- This spacing is ideal because it helps the radiated waves from the two patches add constructively in the main lobe direction, enhancing the overall gain without creating unwanted grating lobes, also increasing isolation between the patches.

Variable	Value	Value In terms of λ_0
D(edge to edge)	24.6 mm	$0.2 \lambda_0$
S(centre to centre)	71.4mm	$0.56 \lambda_0$

Table 1 : Spacing between the antennas

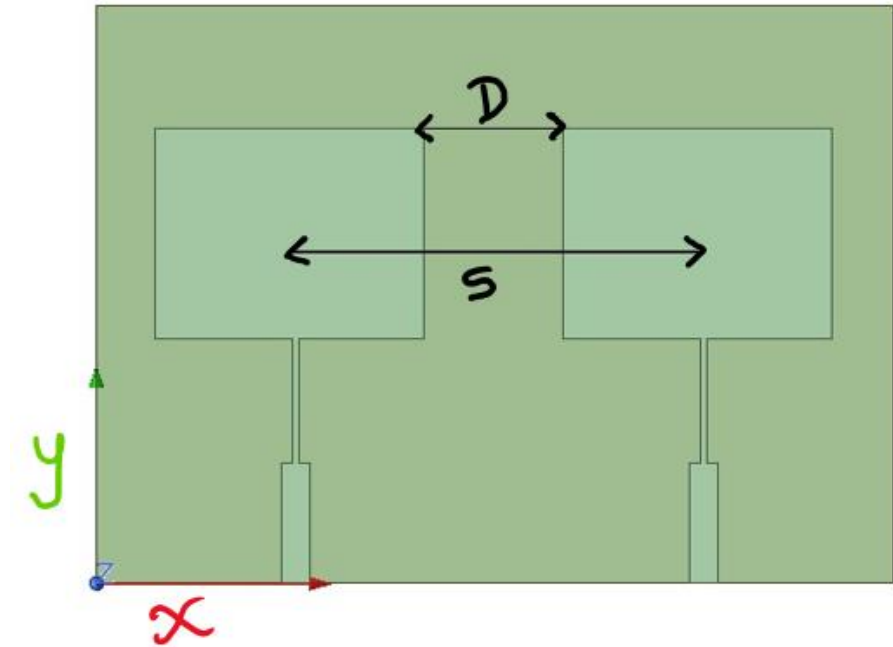


Figure 3.1×2 Patch antenna Array

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.56 \lambda_0$.

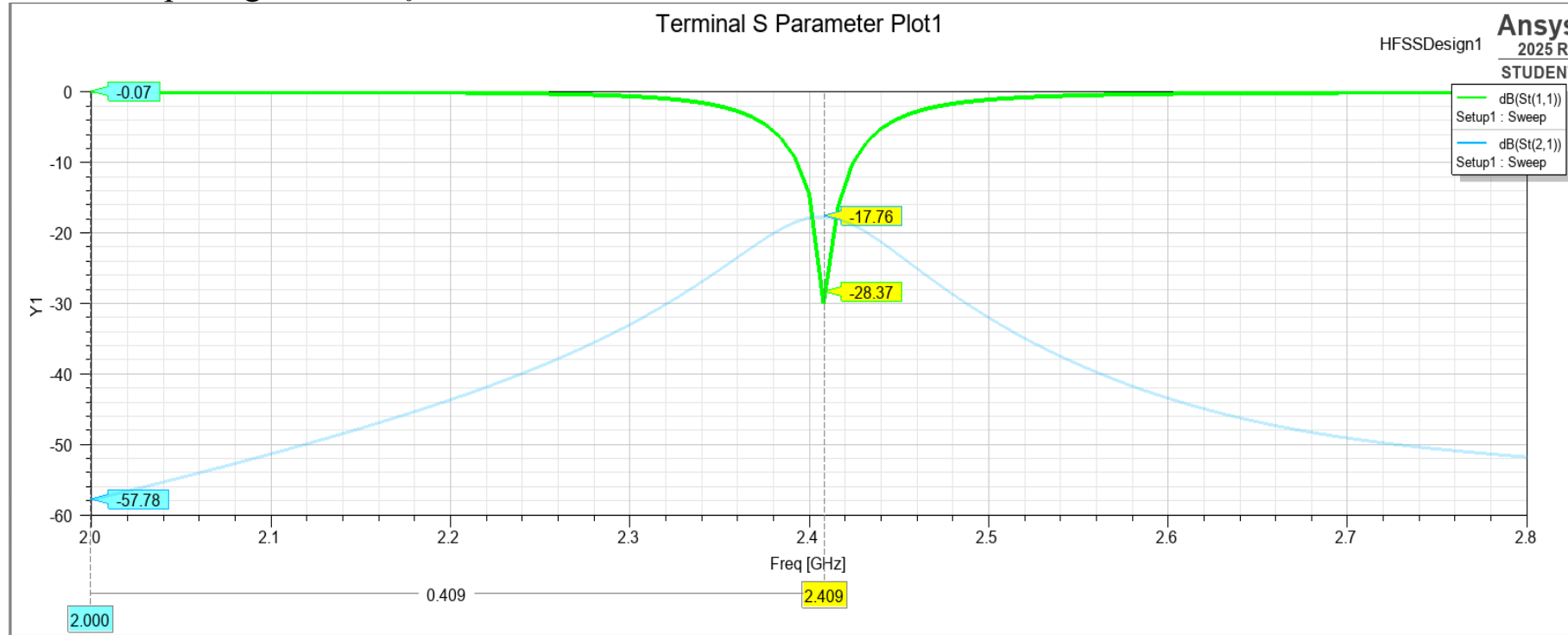


Figure 4. Simulated S_{11} of 1×2 Patch antenna Array

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.56 \lambda_0$.

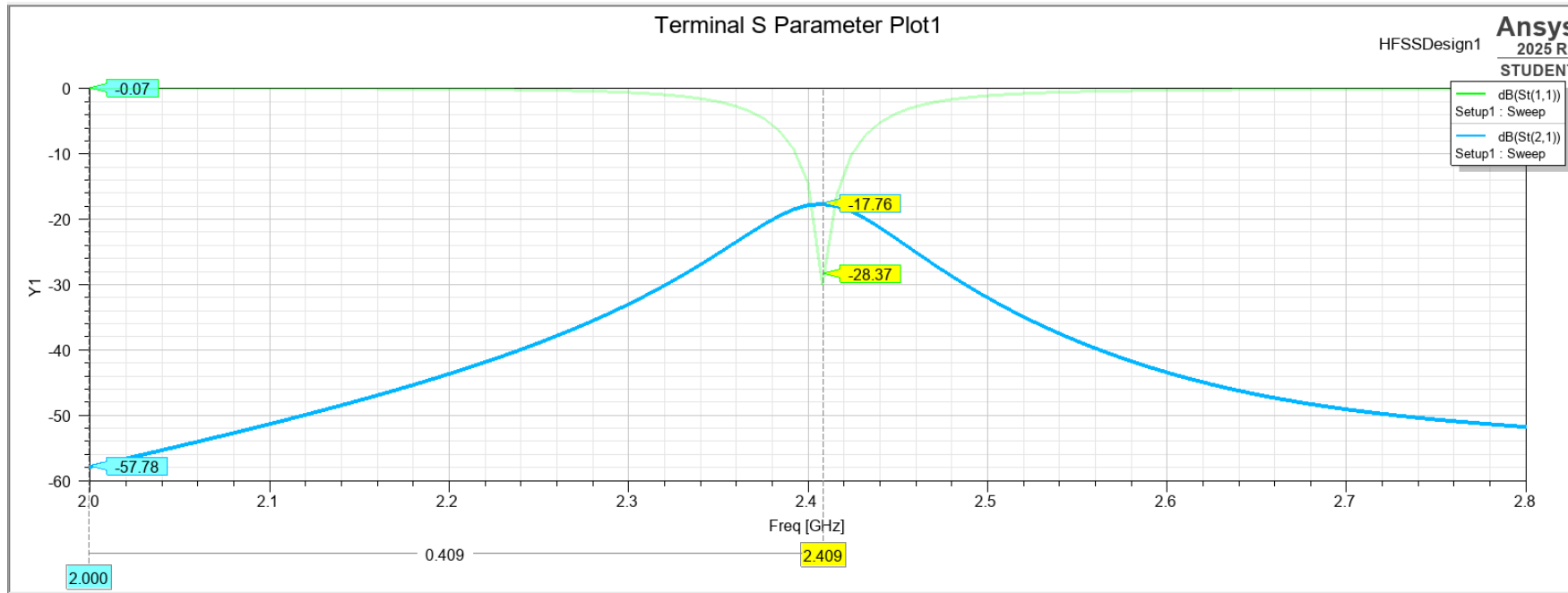


Figure 5. Simulated S_{21} of 1×2 Patch antenna Array

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.5 \lambda_0$.

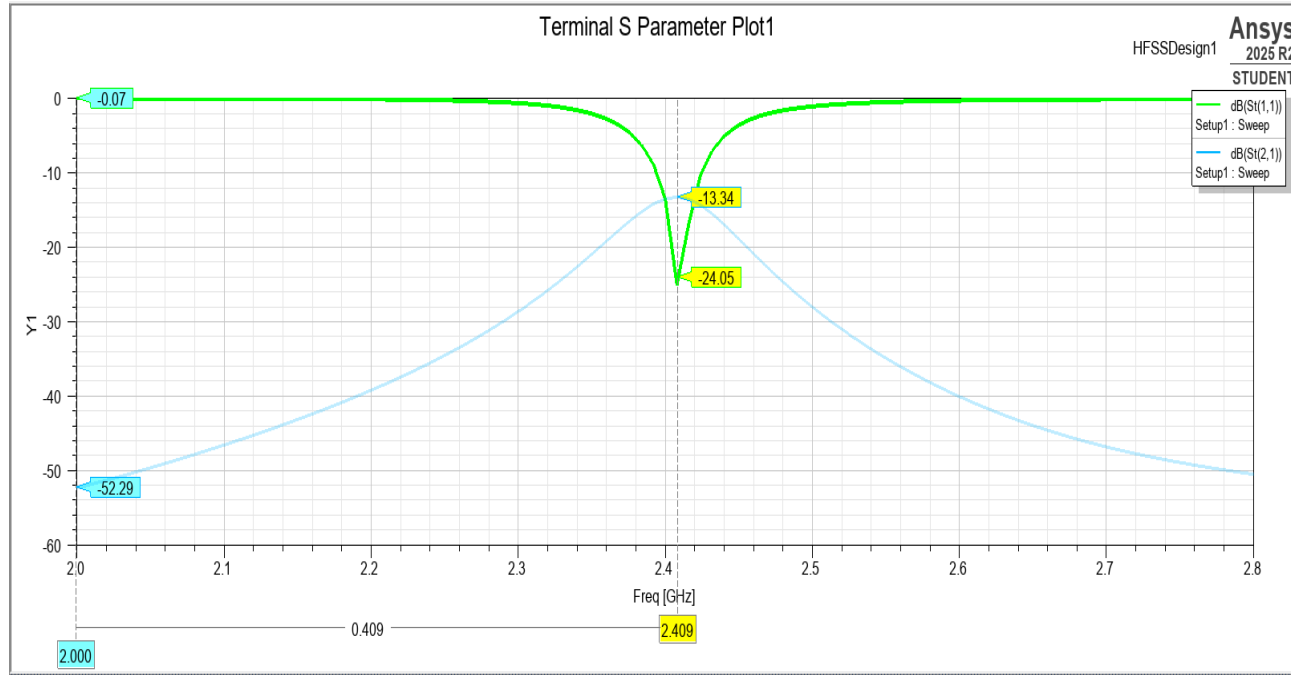


Figure 6. Simulated S_{11} of 1×2 Patch antenna Array

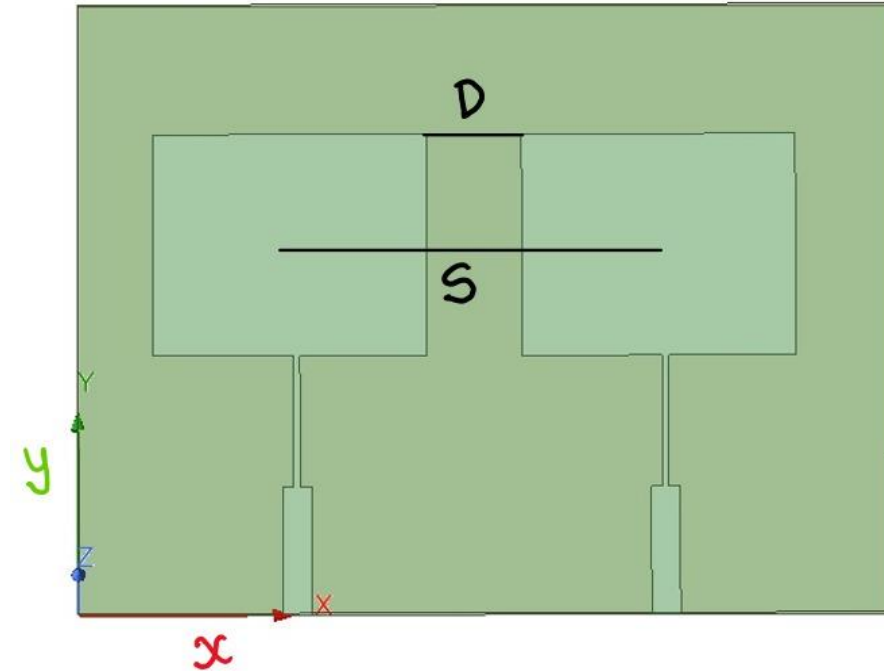
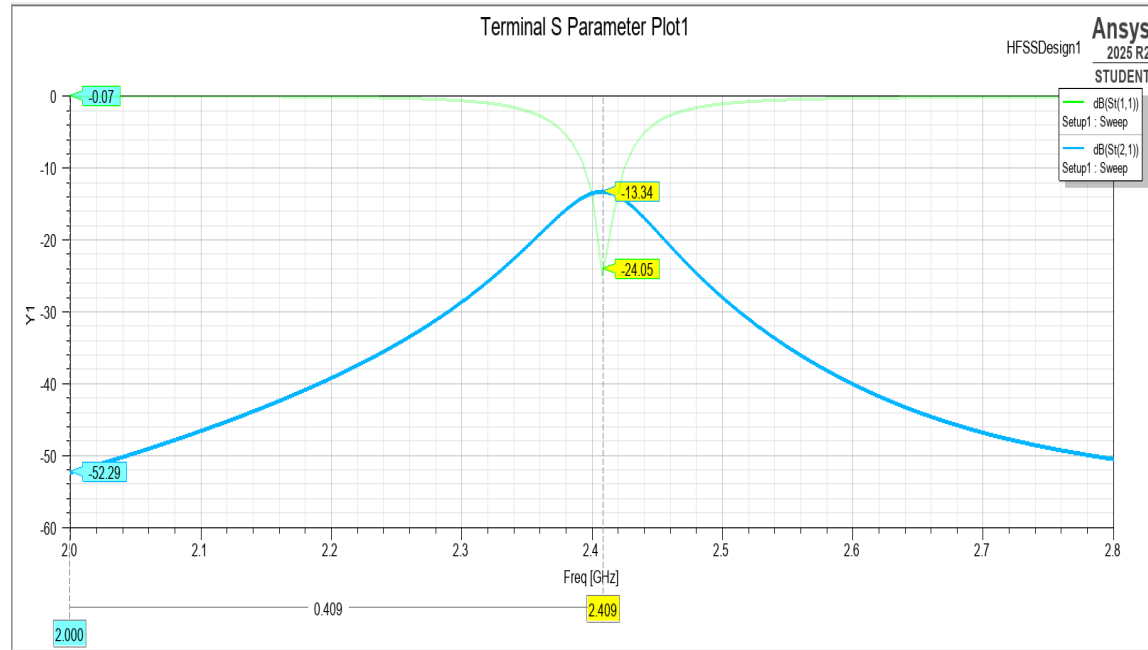


Figure 7. 1×2 Patch antenna Array with centre to centre spacing of $0.5 \lambda_0$.

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.5 \lambda_0$.



Variable	Value	Value In terms of λ_0
D(edge to edge)	17.5 mm	$0.14 \lambda_0$
S(centre to centre)	62.5 mm	$0.5 \lambda_0$

Table 2 : Spacing between the antennas

Figure 8. Simulated S_{21} of 1×2 Patch antenna Array

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.45 \lambda_0$.

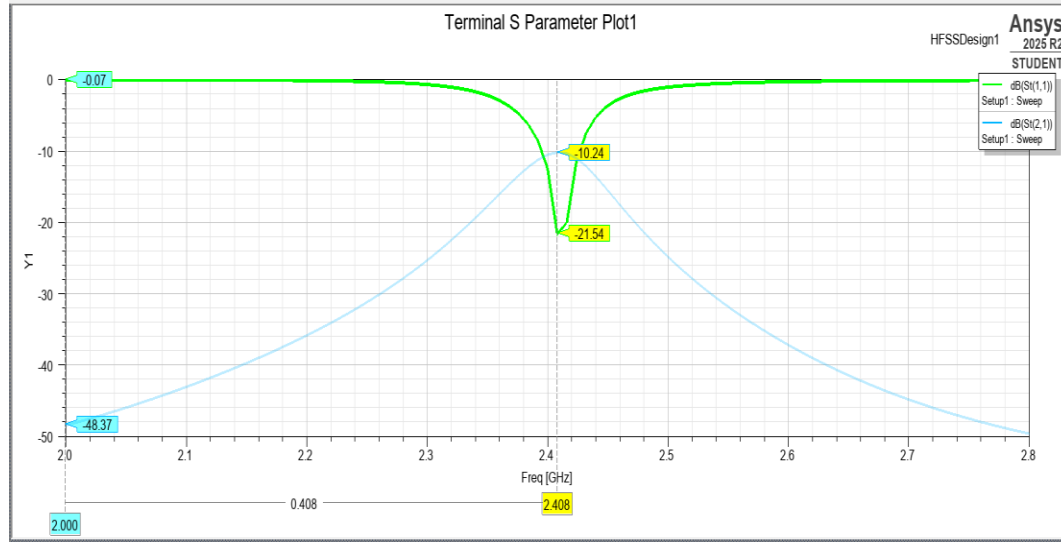


Figure 8. Simulated S_{11} of 1×2 Patch antenna Array

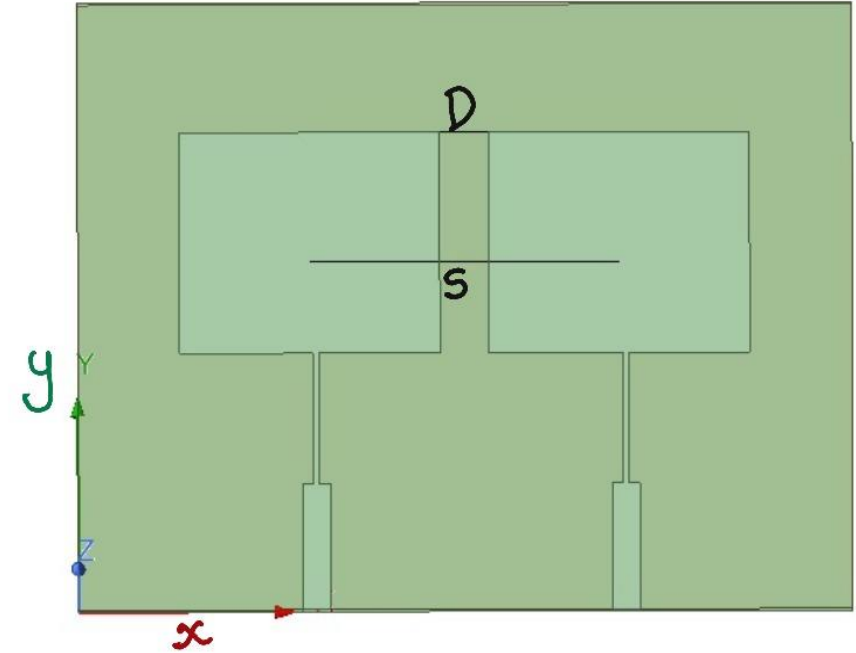
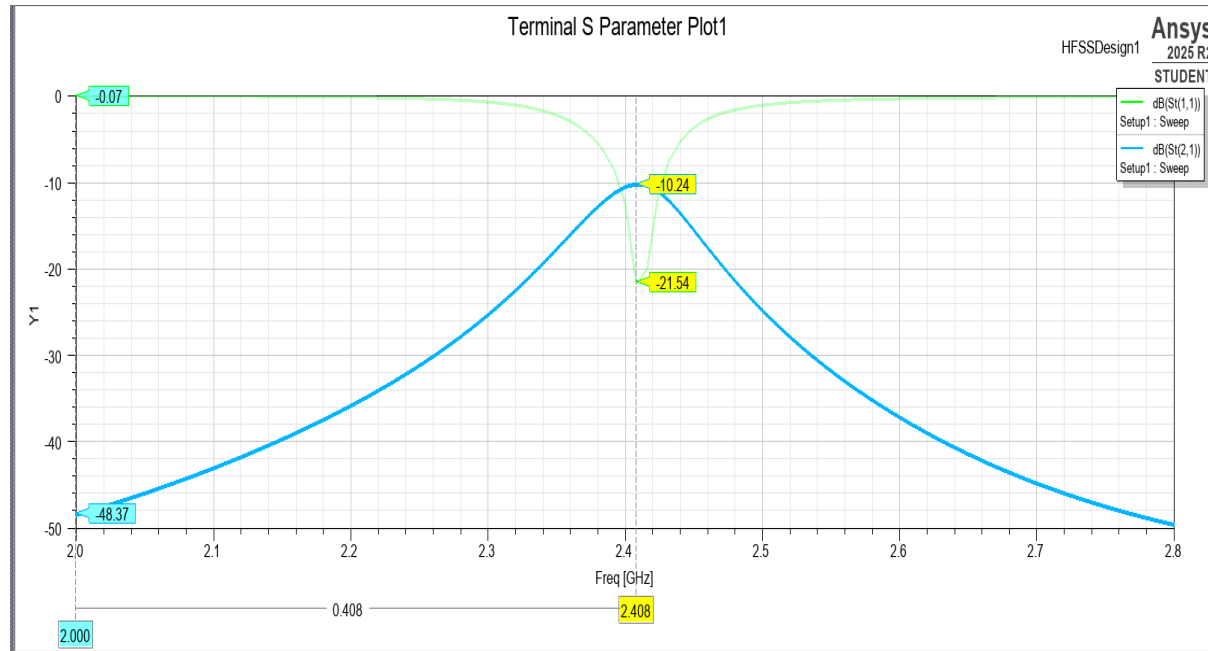


Figure 9. 1×2 Patch antenna Array with centre to centre spacing of $0.45 \lambda_0$.

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.45 \lambda_0$.



Variable	Value	Value In terms of λ_0
D(edge to edge)	11.25 mm	$0.09 \lambda_0$
S(centre to centre)	56.25 mm	$0.45 \lambda_0$

Table 3 : Spacing between the antennas

Figure 10. Simulated S_{21} of 1×2 Patch antenna Array

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.4 \lambda_0$.

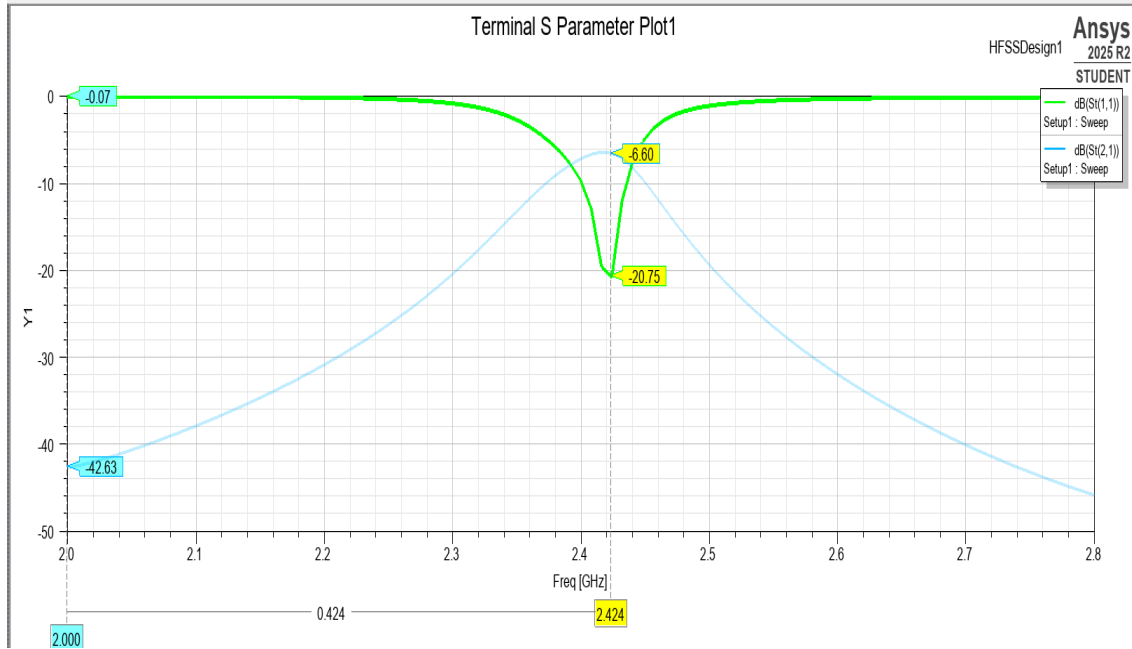


Figure 11. Simulated S_{11} of 1×2 Patch antenna Array

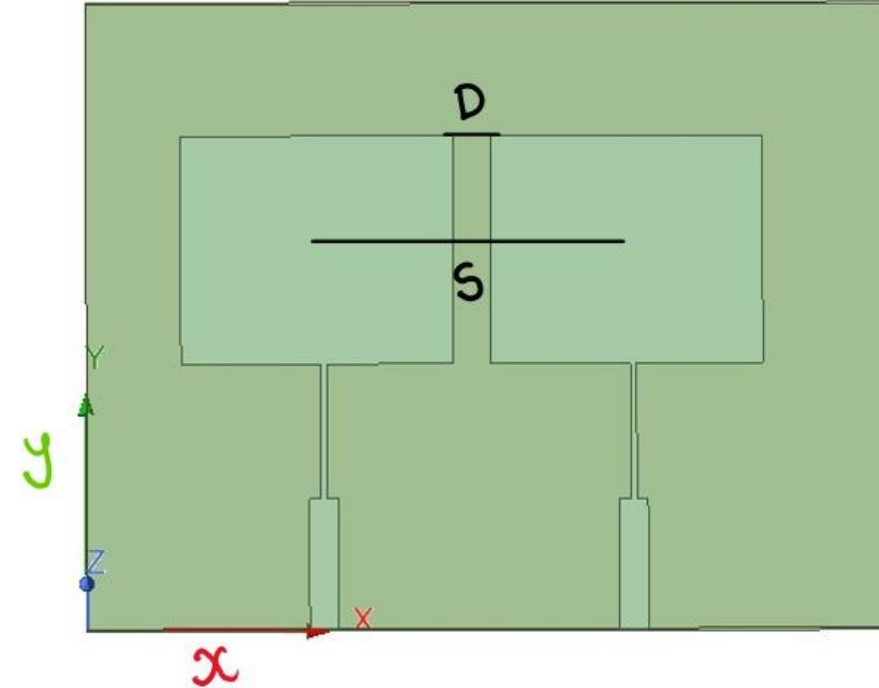
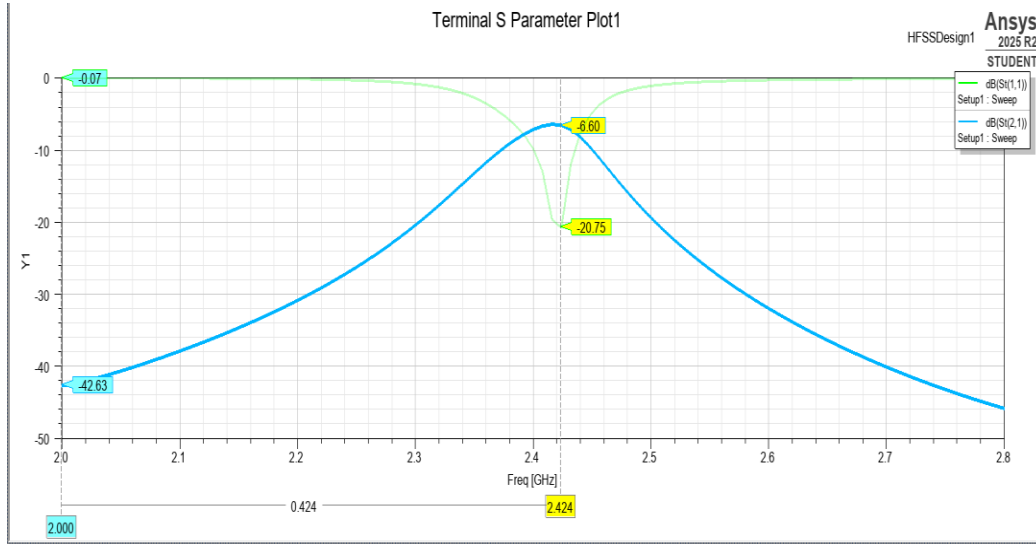


Figure 11. 1×2 Patch antenna Array with centre-to-centre spacing as $0.4 \lambda_0$.

Results:

- Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.4 \lambda_0$.



Variable	Value	Value In terms of λ_0
D(edge to edge)	9.4 mm	$0.04 \lambda_0$
S(centre to centre)	50 mm	$0.4 \lambda_0$

Table 4 : Spacing between the antennas

Figure 12. Simulated S_{21} of 1×2 Patch antenna Array

Results:

Value (centre to centre spacing between antennas)(in mm)	Value In terms of λ_0 (centre to centre spacing between antennas)	Corresponding S_{21}(in dB)
71.4	0.56	17.76
62.5	0.5	13.34
56.25	0.45	10.24
50	0.4	6.6

Table 5. S_{21} for different antenna spacing

Method for Mutual Coupling Reduction :

- Neutralisation line:

A neutralisation line is a **metallic microstrip conductor or strip** introduced between two antenna elements' feed lines or radiators. It creates an artificial coupling path engineered to carry a current that is **equal in magnitude but opposite in phase** to the unwanted mutual coupling current between the elements. This leads to **destructive interference**, reducing the net coupled energy and improving isolation between ports.

- Working: When two antennas are excited, their near fields and surface currents interact and induce a coupling current. The induced current in the NL has **opposite phase** to the naturally coupled current, thus cancelling it at the receiving port.

Design :

- **Length:**

To produce a cancellation current with a suitable phase, the NL is usually designed to be near a **half of wavelength** at the operating frequency. This ensures that the current on the NL arrives at the adjacent port in roughly **180° phase difference** relative to the directly coupled current.

- **Width :**

Width affects the coupling strength — wider lines increase coupling but may disturb matching.

Design:

$$L1+L2+L3+L4+L5+L6+L7+L8+L9+L10 = 62.5\text{mm} = \lambda/2$$

Width=0.25mm

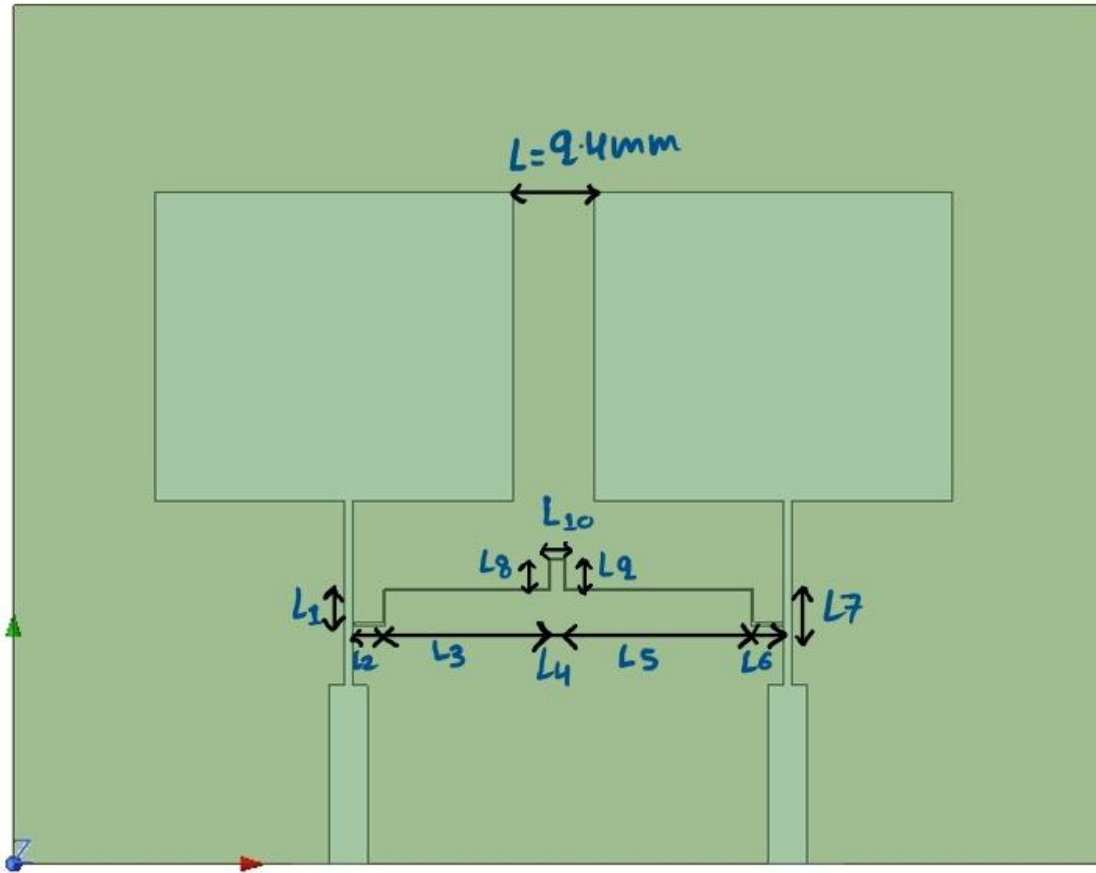


Figure 13. Design of 1×2 Patch antenna Array

Results:

Analysis of the S-parameter plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.4 \lambda_0$.

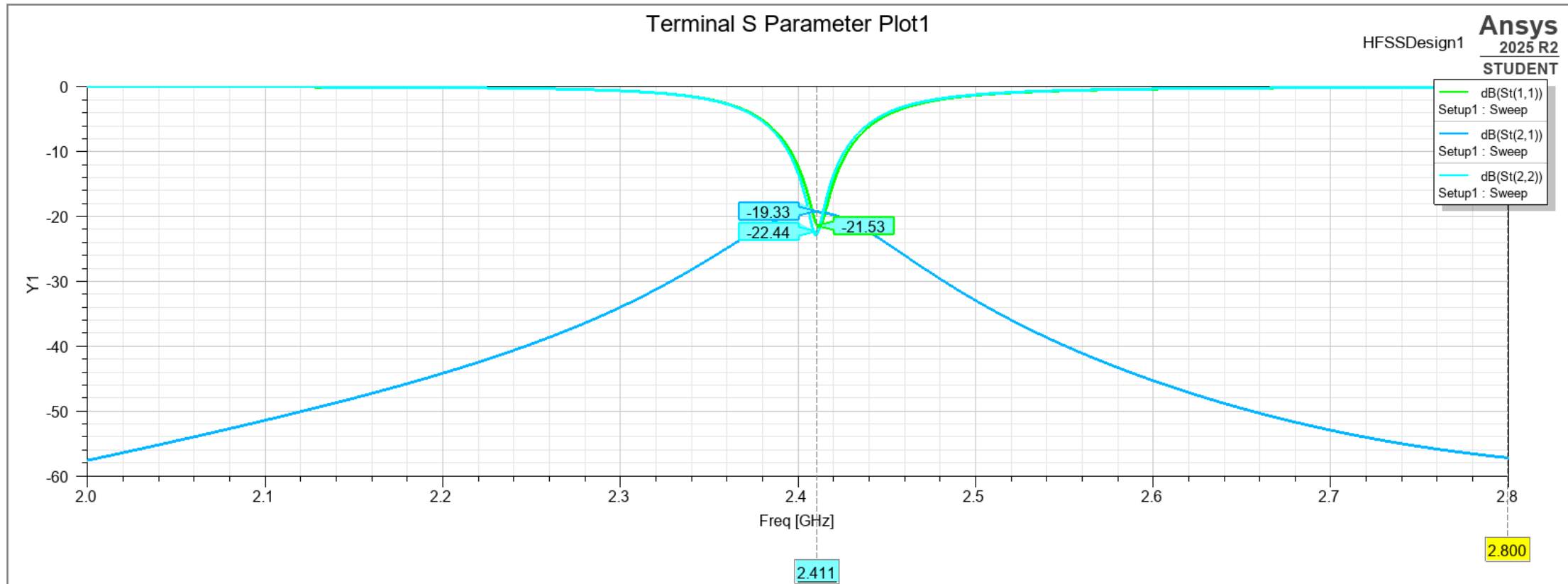


Figure 14. Simulated S parameter of 1×2 Patch antenna Array

Results:

Analysis of the Gain plot for 1×2 Patch antenna Array without a Decoupling Structure with HFSS with centre to centre spacing as $0.4 \lambda_0$.

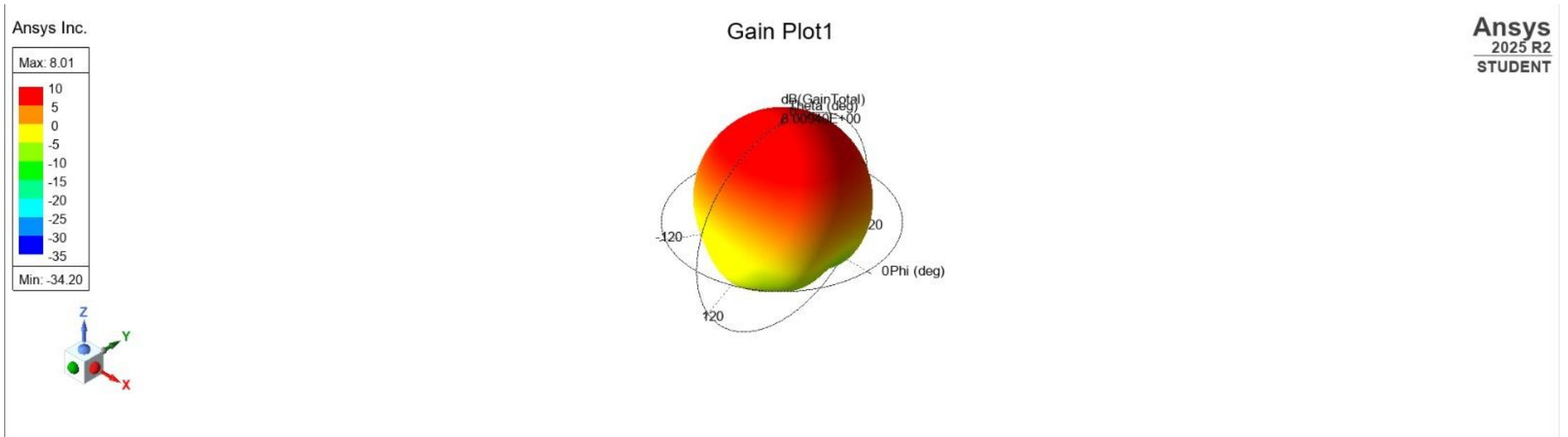


Figure 15. Simulated Gain of 1×2 Patch antenna Array

REFERENCES:

- [1] S.Zhang and G.F.Pedersen, “Mutual coupling reduction for UWB MIMO antennas with a wideband neutralization line,” *IEEE Antennas and Wireless Propagation Letters*, vol.15, pp. 166–169, 2016, doi: 10.1109/LAWP.2015.2438472.
- [2] M. Li, Z. Xu, Y. Ban, C. Sim, and K. Kang, “Isolation enhancement of compact MIMO antennas using neutralization lines,” *IEEE Transactions on Antennas and Propagation*, vol. 67, no. 6, pp. 3850–3859, Jun. 2019, doi: 10.1109/TAP.2019.2903725.
- [3] A. Li, S.-W. Qu, and S. Yang, “Mutual Coupling Reduction of a Dual-Band Four-Port Patch Antenna With Co-Polarized Radiation Pattern by Controlling Electric Fields,” *IEEE Antennas Wireless Propag. Lett.*, vol. 21, no. 9, pp. 1762–1766, Sept. 2022, doi: 10.1109/LAWP.2022.3179404.
- [4] Y. Yang, Y. Feng, L.-K. Zhang, Z. Xing, and J. Li, “A Compact Multifrequency Broadband Dual-Antenna System With Mutual Coupling Reduction for Wi-Fi 6E Application,” *IEEE Antennas Wireless Propag. Lett.*, vol. 23, no. 12, pp. 4074–4078, Dec. 2024, doi: 10.1109/LAWP.2024.3428515.
- [5] Z.-L. Su, K. W. Leung, and K. Lu, “Mutual Coupling Reduction in Dual-Band MIMO Antenna Using Parasitic Dollar-Shaped Structure for Modern Wireless Communication,” *IEEE Trans. Antennas Propag.*, vol. 70, no. 9, pp. 7659–7669, Sept. 2023, doi: [10.1109/TAP.2023.3199567](https://doi.org/10.1109/TAP.2023.3199567).

Thank You